

Size: 165x120 mm

165.00 mm

## Iron (III) Hydroxide Polymaltose Complex

# HAEMO XL CAPSULE

### COMPOSITION:

Each hard gelatin capsule contains:

Iron (III) Hydroxide Polymaltose Complex

Eq. to elemental Iron 100mg

### PHARMACODYNAMICS:

The polynuclear iron (III)-hydroxide cores are superficially surrounded by a number of non-covalently bound polymaltose molecules resulting in an overall complex molecular mass (Mw) of approximately 50 kD, which is so large that diffusion through the membrane of mucosa is about 40 times smaller than that of the hexaqua-iron(II) units.

The complex is stable and does not release ionic iron under physiological conditions. The iron in the poly-nuclear cores is bound in a similar structure as in the case of physiologically occurring ferritin. Due to this similarity, only the iron (III) of the complex is absorbed by an active absorption process. By means of competitive ligand exchange, any iron binding protein in the gastro-intestinal fluid and on the surface of the epithelium, take up iron (III). The absorbed iron is stored mainly in the liver, where it is bound to ferritin. Later in the bone marrow, it is incorporated into haemoglobin.

Iron (III)-Hydroxide Polymaltose Complex has no pro-oxidative properties such as there are in iron II salts. The susceptibility of lipoproteins such as Very Low Density Lipoprotein (VLDL) + Low Density Lipoprotein (LDL) to oxidation is reduced. Iron (III) Hydroxide Polymaltose Complex does not cause teeth staining.

### PHARMACOKINETICS:

Studies using the twin-isotope technique (<sup>55</sup>Fe and <sup>59</sup>Fe) show that

absorption of iron measured as haemoglobin in erythrocytes is inversely proportional to the dose given (the higher the dose, the lower the absorption). There is a statistically negative correlation between the extent of iron deficiency and the amount of iron absorbed (the higher the iron deficiency, the better the absorption). The highest absorption of iron is in the duodenum and jejunum. Iron which is not absorbed is excreted via the faeces. Excretion via the exfoliation of the epithelial cells of the gastro-intestinal tract and the skin as well as perspiration, bile and urine only amount to approximately 1 mg of iron per day. For women, iron loss due to menstruation has also to be taken into account

### INDICATION:

In the treatment of anaemia due to iron deficiency. Treatment and prophylactic therapy of iron deficiency during pregnancy. This product should only be used in pregnancy after the first thirteen weeks.

### DOSAGE AND ADMINISTRATION

Adults: 1 to 2 capsules daily depending upon the severity of anaemia. Medical advice should be sought if symptoms do not improve after four weeks of use of this product as these symptoms may reflect an underlying disease process.

### WARNINGS & PRECAUTIONS:

Jauhkandaripadacapaiankanak-kanak.

Keep out of reach of children.

Please consult your pharmacist/doctor before taking this product.

Contents irritating to mucosa, to ask the doctor.

1. The response to iron therapy should be regularly monitored.
2. The additional requirements for folic acid should be borne in mind when treatment with iron is carried out during pregnancy.
3. In cases of anaemia due to infection or malignancy, the substituted iron is stored in the reticulo-endothelial system, from which it is mobilised and utilised only after curing the primary disease.

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